

Lóránt Nagy, PhD

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Professional Experience

Eötvös Loránd University, ELTE AI Research Group

2023–Present

Research Assistant

- Conduct research on machine-learning methods for stochastic processes and long-memory time series.
- Conducted an interpretability study of neural-network-based Hurst estimation, providing evidence that the model relies primarily on spectral rather than geometric information.
- Develop reinforcement-learning methods for trading under market frictions, comparing learned performance with theoretical asymptotic behaviour.

Alfréd Rényi Institute of Mathematics

2019–Present

Research Fellow

- More recent work focuses on generative AI and deep learning, particularly diffusion-based generative models.
- Developed a framework for studying superlinearly mean-reverting diffusions as alternative forward processes in generative modelling, with ongoing numerical experiments comparing them with the classical dynamics.
- Conduct research in mathematical finance, stochastic control, market frictions, and optimal investment.

Artificial Intelligence National Laboratory

2024

ML Researcher

- Worked with ALTEO on intraday electricity-price forecasting.
- Developed and evaluated machine-learning regression models and assessed their reliability.
- Communicated modelling choices, limitations, and findings to academic and industry collaborators.

Sysler Kft.

2023

Research Consultant

- Prototyped a probabilistic late-fusion framework combining outputs from independent medical diagnostic models, based on Bayesian updating to infer disease progression from empirical patient trajectories.

Morgan Stanley

2017

Interest Rate Modelling Intern

- Applied quantitative methods to problems in interest-rate modelling.
- Gained experience with mathematical modelling and numerical analysis in an industry setting, as well as communicating technical work within a professional team.

Education

Central European University

2018–2022

PhD in Mathematics

Budapest University of Technology and Economics

2015–2018

MSc in Mathematics

Technical Skills

Programming and development tools: Python, PyTorch, scikit-learn, NumPy, SciPy, Git, Docker, GitHub Actions, Wolfram Mathematica, LLM-assisted development workflows.

Quantitative methods: stochastic modelling, simulation, optimization, statistical analysis, mathematical finance, and time-series forecasting.

Cloud and deployment: Azure Static Web Apps, Azure Container Apps; basic CI/CD workflows and containerized application deployment.

Machine Learning and Deep Learning tools: MLPs, RNNs; conditional VAEs, diffusion-based generative models, reinforcement-learning techniques; Weights and Biases, and TensorBoard.

Selected Publications

2026 Rásonyi, M. and Nagy, L.

Optimal investment with transient price impact and stochastic spread.

Preprint, arXiv:2511.12093.

2024 Boros, D., Csanády, B., Ivkovic, I., Nagy, L., Lukács, A., and Márkus, L.

Deep learning the Hurst parameter of linear fractional processes and assessing its reliability.

Quality and Reliability Engineering International, 40, 4228–4246.

2022 Guasoni, P, Nagy, L, Rásonyi, M.

Young, timid, and risk takers.

Mathematical Finance. 31: 1332–1356.

2021 Rásonyi, M. and Nagy, L.

Optimal long-term investment in illiquid markets when prices have negative memory.

Electronic Communications in Probability, 26.

Selected Talks

2026 **Bologna** XIII Bachelier World Congress

On the utility problem in a market where price impact is transient

2025 **Vienna** Vienna Congress on Mathematical Finance, TU Wien

Non-convex optimal investment with transient price impact

2024 **Szeged** Conference on Analysis and Applied Mathematics, University of Szeged

Optimal execution in superlinearly mean-reverting environments

2023 **Budapest** ELTE–Rényi AI Workshop, Eötvös Loránd University

Risk-averse trading

2022 **Ann Arbor** Financial Mathematics Seminar, University of Michigan

Young, Timid, and Risk Takers

Additional Experience and Distinctions

2025 Intermediate-level Qualification in Copyright Law, Hungarian Intellectual Property Office.

2022 Award for Advanced Doctoral Students.

2019 Instructor, *Fundamentals of Stochastic Analysis*, Central European University.

2016 Second Place, TDK competition, *Examples in the theory of continuous trading with friction.*